

Unit I – Learning Problems – Perspectives and Issues - Concept Learning - Version Spaces & Candidate Elimination - Inductive Bias - Decision Tree Learning - Representation Algorithm - Heuristic Space Search

Rubrics (Applicable to all Assignments)

Criterion	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Poor (1)
Problem Understanding	Clearly identifies problem and objectives	Mostly clear	Adequate	Limited understanding	Incorrect understanding
Data Collection & Preparation	Appropriate, clean dataset with preprocessing	Minor preprocessing issues	Basic preprocessing	Poor preprocessing	No preprocessing
Analysis using NumPy/Pandas	Correct and efficient analysis	Mostly correct	Basic analysis	Limited analysis	Incorrect analysis
Visualization & Interpretation	Appropriate graphs with meaningful insights	Good visualization	Basic charts	Minimal interpretation	No visualization
Innovation & SDG Mapping	Excellent real-world relevance with SDG linkage	Good linkage	Moderate linkage	Weak linkage	No SDG relevance
Documentation & Conclusion	Professional report with justified conclusions	Good report	Adequate report	Poor report	Incomplete

Total = 30 Marks

Unit I Analytical Assignments with SDGs

A	Assignment Question	Related SDG
1	Explain how Machine Learning can identify poverty-prone regions using socio-economic data.	SDG 1 – No Poverty
2	Design a Machine Learning model to identify eligible beneficiaries for government welfare schemes.	SDG 1 – No Poverty
3	Explain how Machine Learning can predict crop yield using historical agricultural data.	SDG 2 – Zero Hunger
4	Develop a Machine Learning solution for crop disease detection using leaf images.	SDG 2 – Zero Hunger
5	Explain how Machine Learning can reduce food wastage in supermarkets.	SDG 2 – Zero Hunger

1. Explain how machine learning can identify poverty-prone regions using Socio-economic data

Machine Learning (ML) can identify poverty-prone regions by analyzing socio-economic data such as income, education, employment, housing, healthcare and access to clean water.

Working:

1. collect data about income, education, employment and living conditions.
2. Clean and process the collected data.
3. Apply ML algorithms to identify patterns in the data.
4. Predict which regions are more likely to experience poverty.
5. Display the results using maps or graphs.

Example:- ML can identify villages with low income, poor education and limited healthcare facilities so that the government can provide financial support and basic services.

Example:- ML can identify villages with low income, poor education and limited healthcare facilities so the government can provide financial support and basic service.

Benefits:

- Identifies poverty-prone areas
- Helps the government allocate resources
- Supports better planning and decision-making.

Related SDG; SDG1 - No Poverty

2) Design a Machine Learning model to identify eligible beneficiaries for government welfare schemes.

A Machine Learning model can help the government identify people who are eligible for welfare schemes based on their socio-economic conditions.

Working:

1. Collect data such as family income, family size, employment, age and housing conditions.
2. Preprocess the data by removing errors and handling missing values.
3. Train as ML model using previously verified eligibility records.
4. Use algorithms such as Decision Tree or Random Tree to predict eligibility.
5. Verify the predictions with official rules and human review before approving benefits.

Example:- An ML Model can help identify low-income families who may qualify for scholarships, food assistance or housing schemes.

Benefits:-

- Save time in identifying eligible people.
 - Reduces manual workload.
 - Helps distribute benefits efficiently.
 - Can reduce errors when combined with human verification.
- Related SDG: SDG 1 - NO poverty.

3. Explain how machine learning can predict crop yield using historical agricultural data.

Machine learning can predict crop yield by analyzing historical agricultural data such as rainfall, temperature, soil fertility, irrigation and fertilizer usage.

Working:

1. Collect past agricultural data, including crop production, rainfall and soil conditions.
2. Clean and organize the data.
3. Train an ML model using algorithms such as Linear Regression or Random Forest.
4. Provide current weather, soil and farming data to the trained model.
5. Predict the expected crop yield.

Example: ML can predict the expected yield of rice based on rainfall, temperature, soil quality and previous year's production.

Benefits:-

- Helps farmers plan agricultural activities.
- Improves resource management.
- Supports better crop planning
- Helps reduce unexpected crop losses.

Related SDG: SDG 2 - Zero Hunger.

4. Develop a Machine Learning solution for crop diseases detection using leaf images.

Machine Learning, especially Deep Learning, can detect crop diseases by analyzing images of plant leaves.

Working:

1. collect images of healthy and diseased plant leaves.
2. Label the images according to the diseases
3. preprocess the images by resizing and normalizing them.
4. Train a Convolutional Neural Network (CNN) to recognize disease patterns.
5. Upload a new leaf image to the trained model.
6. The model predicts whether the leaf is healthy or diseased and identifies the likely disease.

Example:- A farmer can take a picture of a tomato leaf. The ML model analyses the image and predicts whether the plant has a disease such as Early blight.

Benefits:-

- Detects diseases at an early stage
- Helps farmers take timely action
- Reduces crop losses
- Saves time and effort.

Related SDG : SDG 2 - Zero Hunger.

5) Explain how machine learning can reduce food wastage in supermarkets.

Machine learning can reduce food wastage in supermarkets by predicting customer demand, managing inventory, and identifying products that may remain unsold or expire soon.

Working:

1. Collect data about previous sales, customer demand, seasonal trends and product expiry dates.
2. Analyze the data using ML algorithms.
3. Predict the demand for different products.
4. Recommend suitable stock levels and identify items that need attention before expiry.
5. Help supermarkets plan discounts and stock replenishment.

Example: ML can predict that a supermarket will need fewer bakery products on a particular day. The supermarket can adjust its stock to avoid producing or ordering excess items.

Benefits-

- Reduces food wastage
- Improves inventory management
- Saves money
- Helps ensure food availability

Related SDG : SDG 2 - Zero Hunger.